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REVIEW

Impact of bariatric surgery on oral health: a systematic review and meta-analysis

Aline Xavier Ferraz[1,2] · Flavio Magno Gonçalves[2,3] · Pedro Duarte Ferreira-Neto[2,4] · Rosane Sampaio Santos[2,3] · Odilon Guariza-Filho[2,5] · Bianca Simone Zeigelboim[2,3] · Camila de Castro Corrêa[2,6] · Karinna Veríssimo Meira Taveira[2,7] · Cristiano Miranda de Araujo[2,3]

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Abstract

Objective To evaluatethe impact of bariatric surgery (BS) on the oral health status of obese individuals.

Material and methods The search was performed on the Cochrane Library, Embase, Latin American and Caribbean Literature on Health Sciences (LILACS), LIVIVO, PubMed/Medline, Scopus, and Web of Science databases. Grey literature was also consulted through Google Scholar, OpenGrey, ProQuest, and MedRxiv. The risk of bias was assessed using the Joanna Briggs Institute Critical Appraisal tool. Pre- and post-surgical moments were compared through random effects meta-analysis. The GRADE (Grading of Recommendations, Assessment, Development and Evaluations) assessment tool was used to judge the certainty of evidence.

Results After searching the databases, 908 references were retrieved, with 30 articles selected for synthesis. When comparing the pre- and postoperative moments, there was no difference in the index of decayed, missing, or filled teeth. Furthermore, salivary flow and probing depth presented a slight increase. Post-surgery patients showed an increase (%) in tooth wear at the dentin level [MD = -6.23; IC95% = -8.45—-4.01; $I^2 = 0\%$].

Conclusion Patients undergoing BS show no difference when considering the rate of caries or missing teeth and little to no effect was observed on salivary flow rates and periodontal probing depth. On the other hand, greater attention should be given to dentin wear in post-surgical patients of BS.

Clinical relevance Patients undergoing BS should receive careful monitoring regarding oral health by doctors, dentists, and the entire multidisciplinary team involved before and after the surgical procedure.

Keywords Obesity · Bariatric surgery · Oral health · Tooth wear · Systematic review

✉ Cristiano Miranda de Araujo
cristiano.araujo@utp.br

Aline Xavier Ferraz
alinexferraz@gmail.com

Flavio Magno Gonçalves
flaviomagno93@yahoo.com.br

Pedro Duarte Ferreira-Neto
pedroduartef25@gmail.com

Rosane Sampaio Santos
rosane.santos2@utp.br

Odilon Guariza-Filho
odilongfilho@gmail.com

Bianca Simone Zeigelboim
bianca.zeigelboim@utp.br

Camila de Castro Corrêa
camila.ccorrea@hotmail.com

Karinna Veríssimo Meira Taveira
karinnavm@hotmail.com

1 Undergraduate Dentistry Program, Tuiuti University of Paraná, Curitiba, Brazil

2 Center for Advanced Studies in Systematic Review and Meta-analysis (NARSM), Curitiba, Paraná, Brazil

3 Postgraduate Program in Communication Disorders, Tuiuti University of Paraná, Curitiba, Brazil

4 Undergraduate Department of Dentistry, Federal University of Rio Grande Do Norte, Natal, Rio Grande Do Norte, Brazil

5 Department of Orthodontics, Pontifícia Universidade Católica Do Paraná, Curitiba, Paraná, Brazil

6 Department of Speech Therapy and Audiology, Centro Universitário Planalto Do Distrito Federal, Brasília, Distrito Federal, Brazil

7 Department of Morphology- Center of Biosciences, Federal University of Rio Grande Do Norte, Natal, Rio Grande Do Norte, Brazil

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

Introduction

Obesity is a chronic disease [1] that affects more than 650 million adults worldwide [2, 3], according to the World Health Organization (WHO). Its origin is multifactorial, being characterized by excess fat and associated factors, such as changes in metabolism, genetics, and a sedentary lifestyle, as well as behavioral, psychological, social, and dietary factors [1, 4].

In recent decades, the search for procedures that result in weight loss has been increasing [1, 5]. Bariatric surgery (BS) is a set of techniques whose main objective is weight reduction and the regression of correlated diseases, such as hypertension, type 2 diabetes mellitus, arteriosclerosis, arthritis, hyperlipidemia, sleep apnea syndrome, and endocrine dysfunctions [1, 4, 6, 7].

There are reports in the literature showing that patients undergoing BS can acquire some oral disorders, such as dental erosion, which is caused by the decrease in the salivary pH and consequent dissolution of mineralized tissue [4]; dental hypersensitivity; acid reflux and vomiting; active dental caries; chewing difficulty; hyposalivation; xerostomia; periodontal disease [1, 4, 8]; functional limitation; oral physical pain; and tooth loss [9].

Other reviews evaluated the impact of BS on isolated aspects of oral health status, such as on dental erosion [10], salivary flow [11], periodontal status [2, 11–13], and caries [14]. However, new primary studies were published and there are still no reviews quantifying the caries index, missing teeth, and tooth wear in this population through meta-analyses. Furthermore, there is a need for broader search strategies using more databases to decrease the likelihood of publication bias.

Thus, this systematic review aims to evaluate the impact of bariatric surgery on the oral health status of obese individuals.

Material and methods

This systematic review was developed in accordance with the *Preferred Reporting Items for Systematic reviews and Meta-Analyses* (PRISMA 2020 Statement) [15].

Protocol and registration

The protocol was registered in the PROSPERO website (International Prospective Register of Systematic Reviews – York University Center for Commentary and Dissemination).

Eligibility criteria

The acronym PICOS was used to establish the eligibility criteria for the following question: what is the impact of bariatric surgery on the oral health of obese individuals?:

- P (population) = obese patients of both genders;
- I (intervention) = bariatric surgery;
- C (comparison) = pre- and postoperative period (paired or unpaired);
- O (outcomes) = the impact on oral health status regarding dental aspects, periodontal aspects, masticatory function, flow, and salivary composition;
- S (study design) = Randomized, non-randomized, quasi-experimental or quasi-randomized controlled trials, cohort, case–control or cross-sectional studies;

Inclusion criteria

Studies addressing a population of adult individuals (> 18 – 60 years of age) with an indication for BS were included. The studies should have compared pre- and postoperative moments (paired or unpaired) and should have evaluated at least one of the following aspects: dental aspects, periodontal aspects, masticatory function, flow, and salivary composition. Randomized, non-randomized, quasi-experimental or pseudo-randomized clinical trials, as well as observational cohort, case–control or cross-sectional studies were considered eligible. No studies were excluded based on gender, ethnicity, language or time of publication.

Exclusion criteria

1. Studies in which BS was not performed or was associated with another type of surgery;
2. Studies in which there was no comparison between at least two evaluation moments;
3. Studies that have not evaluated at least one of the outcomes of interest or with missing or incomplete data even after trying to contact the authors;
4. Reviews, expert opinions, in vitro or animal studies, letters, conference abstracts, case reports, and case series.

Information sources and search strategy

Appropriate word combinations and truncations were performed and adjusted for each of the following electronic databases: Cochrane Library, Embase, Latin American and Caribbean Literature on Health Sciences (LILACS), LIVIVO, PubMed/Medline, Scopus, and Web of Science (Appendix 1). The additional literature search included grey

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

literature through Google Scholar, Open Grey, ProQuest, and MedRxiv. Database searches were performed on October 10, 2021 and updated on July 4, 2022.

An expert on the topic was consulted to indicate papers that could be included. In addition, a manual search was performed in the references of all included papers to verify papers with potential for inclusion. All references were managed using the appropriate software (EndNote® Web—Thomson Reuters, Philadelphia, PA) and duplicates were removed.

Selection process

The papers were analyzed in two phases. In Phase 1, two reviewers separately reviewed the titles and abstracts of all references. Papers that did not meet the eligibility criteria were excluded. In Phase 2, the same reviewers fully read the selected papers independently. In order to ensure greater agreement between reviewers, a calibration through a partial selection of the literature was performed before the beginning of Phase 1, with the complete reading being initiated only after obtaining > 0.7 in the Kappa Coefficient.

In case of disagreement between the first and second reviewers (in which there was no resolution by debate), a third reviewer was included for a tie-breaking vote.

Data collection process

Data was extracted by two reviewers, who collected information from the included studies. Disagreements were resolved through discussion with a third reviewer. The collected data included study characteristics (authors, year of publication, country, and study design); clinical characteristics of the evaluation (pre- and postoperative period); characteristics of the results (the presented results concerning each study); and conclusions. In the case of incomplete data, three attempts were made to contact the authors (first and last author or corresponding author) to obtain information. The paper was excluded when there was no response.

The collected data were managed by the *Rayyan—Intelligent Systematic Review* software (https://rayyan.qcri.org/). The final selection was based on the full text.

Data items

Mean and standard deviation values of the outcomes of interest were extracted from the included studies. In the absence of these values, mathematical approximation methods were used to estimate these data through the median and interquartile range (when available) [16]. Values extracted from studies were only accepted if obtained through validated questionnaires or clinical measurements. The sample size at each evaluation moment was also extracted from the included papers.

Risk of bias in individual studies

The Joanna Briggs Institute's Critical Appraisal tool [17] was used with the appropriate checklist for each study design to measure the methodological quality of the selected studies. Two reviewers performed the risk of bias assessment separately and judged the included papers by marking each evaluation criterion with "yes", "no", "uncertain", and "not applicable". The risk of bias was classified as high when the study reached 49% "Yes"; moderate when the study reached 50% to 69% "Yes"; and low when the study reached more than 70% "Yes" [18]. Disagreements were resolved through discussion with a third reviewer.

Effect measures

Since the outcomes of interest are reported using continuous data of a same scale, the difference between means (MD) was calculated by comparing baseline values for each outcome with the number of post-surgical follow-up times or with a control group.

For outcomes that showed statistical significance, the effect size was obtained by calculating the Hedges' *g* [19], where values from 0.2 to 0.5 were considered small, 0.5 – 0.8 were considered moderate, and values > 0.8 were considered as large effect sizes [20].

Synthesis methods

The amount of three papers that met the eligibility criteria for carrying out quantitative synthesis and with the necessary data for each outcome was stipulated as the minimum. A method for random effects meta-analysis was determined using the Stata 16.0 (Stata Corp LLC, College Station, USA) statistical software, in which the weight of the studies was weighted by the inverse variance method. Heterogeneity was calculated by the inconsistency index (I^2) and the variance was calculated by Tau^2, which was estimated by the DerSimonian-Laird method. Confidence intervals of 95% were generated (CI95%), with a significance level of 5%.

When the presence of heterogeneity was detected, a subgroup analysis was performed considering possible categorical variables that may have acted as a confounding factor for the analysis.

Reporting bias

Publication bias was assessed using the funnel plot and the Egger test for outcomes with a number > 10 included studies, considering a significance level of 5%. Furthermore, a comprehensive

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

search strategy including several databases, which also included a non-English language database (LILACS) and grey literature was carried out to reduce the likelihood of publication bias.

A sensitivity analysis was performed for studies with a high risk of bias to evaluate the influence of these studies on the generated estimates.

Certainty of evidence

The certainty of evidence for each stipulated outcome was evaluated using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) tool [21]. First, the certainty of evidence was judged considering the following domains: risk of bias, inconsistency, imprecision, indirect evidence, and publication bias. Then, the certainty of evidence for each outcome was judged as 'very low', 'low', 'moderate' or 'high'.

Results

Study selection

The search strategy retrieved a total of 908 references in the six electronic databases. After removing duplicate references, 721 references remained. After reading the titles and abstracts (Phase 1), a total of 31 papers were selected for full reading (Phase 2), among which 20 were excluded (Appendix 2). Fourteen additional papers from a reference list, expert consultation, and grey literature were included. After updating the search, five papers were included in the synthesis, totaling 30 papers (Fig. 1).

Study characteristics

The included papers originated from the following countries: Saudi Arabia, Brazil, Canada, France, Holland, India, Israel, Poland, Portugal, Spain, and Switzerland. The sample comprised 11 cross-sectional studies, 18 cohort studies, and one case–control study. The year of publication ranged between 2011 and 2020. There was a higher prevalence of females in the studies. There was also a comparison between eutrophic, obese, and BS patients. All features of the included studies are available in Table 1.

Risk of bias in the selected studies

Regarding risk of bias, 18 studies presented a low risk of bias [8, 13, 22–37], nine presented a moderate risk of bias

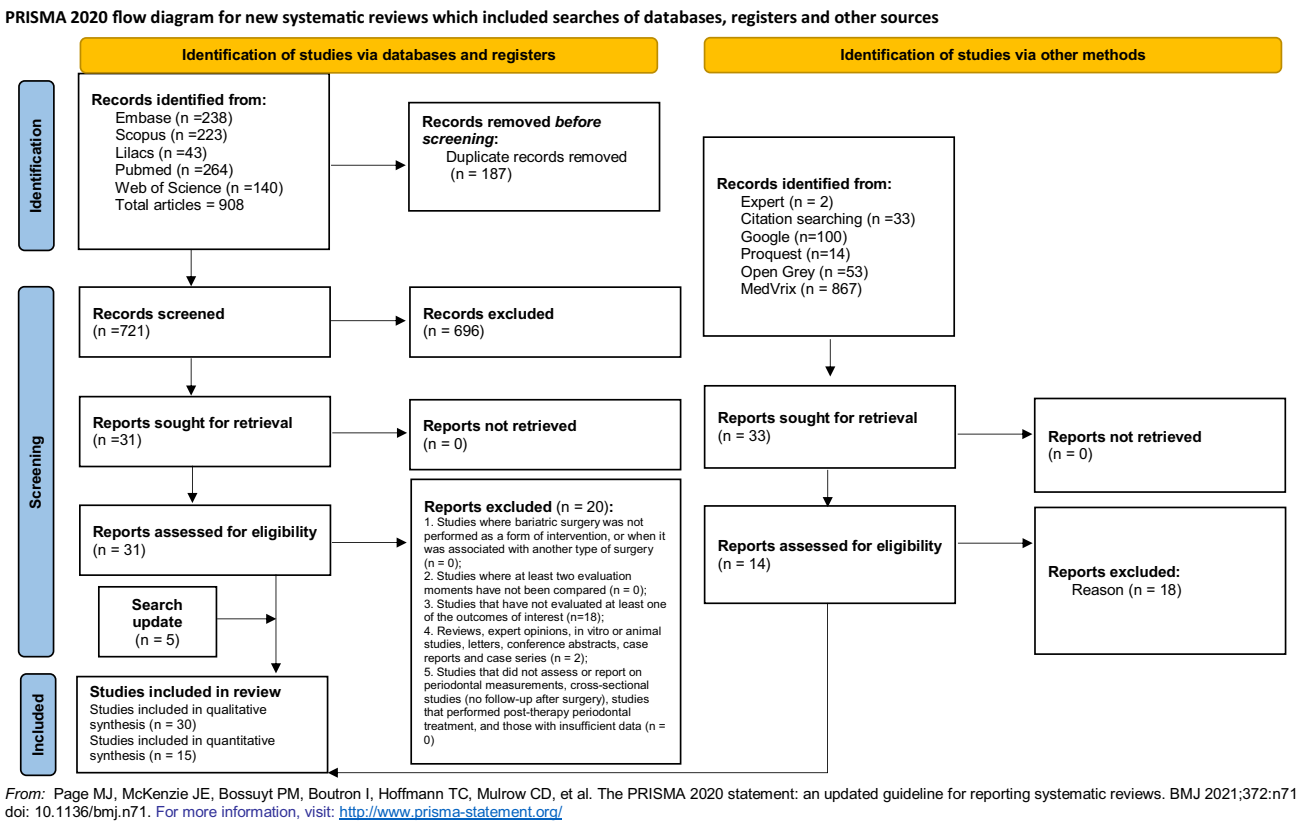

Fig. 1 Flowchart of literature search and selection criteria

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

Table 1 Characteristics of the included studies

Author, Year (Country)	Study design	Sample	Age (Mean or range)	Type of surgery (Postoperative follow-up time)	Conclusions
Alves, M.S.C. et al. 2012 (Brazil)	Cross-sectional	125 patients, being patients who had already undergone bariatric surgery for at least 6 months (bariatric group), morbidly obese patients who were on the waiting list for this surgery (obese group) and patients awaiting outpatient care in other sectors (control group).	16–70y	NR (6 months)	Reflux and vomiting did not seem to influence severity of noncarious dental lesions positively.
Aznar, F.D. et al. 2019 (Brazil)	Cross-sectional	240 individuals, 48 female and 12 men	NR	Roux-en-Y gastric bypass (Up to 24 months, and more than 36 months)	They presented more tooth wear in the incisal / occlusal surfaces, and the teeth were more affected by the anterior ones.
de Moura-Grec P.G. et al. 2012 (Brazil)	Cohort	91 morbidly obese patients, 74 women and 16 men.	38.9y	NR (6 and 12 months)	Bariatric surgery promoted a negative impact on oral health conditions, especially in periodontal disease and tooth wear. Changes in the amount of periodontopathogenic bacteria were observed in the patients evaluated during the three periods.
Džunková, M. et al. 2020 (Spain)	Cohort	35 adult patients with obesity undergoing bariatric surgery, 17 women and 18 men	48y	Gastrectomy vertical, Roux-en-Y gastric bypass, gastric bypass with omega loop and laparoscopic gastric plication (1 day, 3 and 12 months)	There are many factors that can influence the composition of an individual's salivary microbiome, such as dental hygiene, oral health status, nutritional composition of food, immunological factors and also bacteria specific residents of the individual who can prevent colonization by new species. Many of these factors cannot be quantified accurately, or its action on the salivary microbiome is not straightforward. The alternation observed in the microbiome may be related to changes in diet.
Gambus L.C.C. et al. 2018 (Brazil)	Cross-sectional	The population sample of this research was composed of 74 patients, 40 morbidly obese and 34 with body mass within normal limits.	33,4y	NR (NR)	Both weight gain and bariatric surgery are risk factors for the oral condition, causing alteration in some important salivary components, such as total proteins, amylase, calcium and glucose.
Godlewski, A.E. et al. 2011 (France)	Cohort	46 obese women- 3 groups formed. Fully toothed; PD Group: Partially dentate without partial dentures; DW Group: Users of full and partial dentures.	41y	NR (3 months)	Interactions between the dental state and food properties can affect the consequences of changes in chewing activities in the composition of the pre-swallowed bolus.
Hashizume, L.N. et al. 2015 (Brazil)	Cohort	27 morbidly obese patients (BMI > 40 kg/m(2)), before and 6 months after bariatric surgery	over 18y	Roux-en-Y gastric bypass (6 months)	The results suggest that salivary levels of mutans streptococci increased after bariatric surgery in patients with morbid obesity.
Jaiswal G.R. et al. 2015 (India)	Cross-sectional	224 patients of both sexes underwent a complete periodontal examination before bariatric surgery and 6 months after surgery.	18–64y	NR (6 months)	A fibrous diet along with good periodontal care can help improve the state of oral hygiene of patients undergoing bariatric surgery, even if periodontal intervention is not performed resulting in the absence of periodontitis, thus improving the patient's quality of life.

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

Table 1 (continued)

Author, Year (Country)	Study design	Sample	Age (Mean or range)	Type of surgery (Postoperative follow-up time)	Conclusions
Knas, M. et al. 2016 (Poland)	Cohort	40 morbidly obese	34–55y	Gastric sleeve (6 months)	In morbid obesity, reduced unstimulated and stimulated salivary flow can be observed. Bariatric surgery restored only unstimulated salivary flow to normal values.
Lakkis, D. et al. 2011 (Saudi Arabia)	Cohort	30 obese. Of these, 15 patients had already undergone BC and lost ‡40% of their excess weight for ‡6 months after surgery. The other 15 patients were also obese but did not have the surgery or lost weight to serve as a control group.	47,1y	NR (4 to 6 weeks)	An improved response to non-surgical periodontal therapy is observed in obese patients who had significant weight loss after bariatric surgery compared with obese patients who did not have such a surgery.
Lamy, E. et al. 2015 (Portugal)	Cross-sectional	18 women	22–55y	Gastric sleeve (NR)	Results of this study clearly showed a salivary proteome for obese individuals differing from that of individuals with normal weight. In addition, some differences were not observed in the case of those who underwent bariatric surgery. Some proteins with different levels may be related to metabolic changes associated with obesity, while others are salivary proteins that have been referred to as involved in the perception of taste. The findings of this study point to the possible use of saliva in the study of obesity and suggest a possible relationship between salivary proteome and taste sensitivity in obese individuals, and may eventually result in different food intake.
Marsicano, J.A. et al. 2011 (Brazil)	Cohort	54 obese patients undergoing bariatric surgery, 44 women and 10 men	40.5y	Roux-en-Y gastric bypass (3 and 6 months)	Changes in lifestyle after bariatric surgery can increase the severity of pre-existing dental problems. However, these changes in oral health did not influence quality of life.
Marsicano, J.A. et al. 2012 (Brazil)	Cross-sectional	102 patients - Fifty-two patients underwent bariatric surgery (Rux-en-Y gastric bypass)and 50 severely obese patients indicated for bariatric surgery	37.6y	Roux-en-Y gastric bypass (16.9 months)	The prevalence of oral diseases was similar in patients with severe obesity candidates for bariatric surgery and in patients undergoing bariatric surgery. Nevertheless, there was a higher prevalence of periodontal pockets in the operated group.
Moura-Grec, P.G. et al. 2014 (Brazil)	Cohort	90 obese patients for the initial sample, of these, 59 were followed up 6 months after bariatric surgery. 51 non-obese volunteers were also evaluated to compose the control group.	38.9y	Roux-en-Y gastric bypass (6 months)	Bariatric surgery can improve systemic conditions. However, it had a negative impact on oral health conditions due to increased periodontal disease and tooth wear.
Pataro, A.L. et al. 2012 (Brazil)	Cross-sectional	345 participants between 18 and 60 years of age, of both sexes,	18–60y	NR (Up to 6 months, and more than 6 months)	Differences in periodontal condition were observed in individuals at different times of bariatric surgery, showing a high prevalence of periodontitis both in preoperative and postoperative follow-up.

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

Table 1 (continued)

Author, Year (Country)	Study design	Sample	Age (Mean or range)	Type of surgery (Postoperative follow-up time)	Conclusions
Pataro, A.L. et al. 2016 (Brazil)	Cross-sectional	154 patients	18–65y	NR (24 months)	Bariatric surgery showed an inverse microbial effect in the oral and stomach environments, revealing higher oral and stomach bacterial frequencies.
Porcelli, I.C.S. et al. 2016 (Brazil)	Cross-sectional	Twenty-nine patients, of these, 18 accepted to have their dental conditionalso clinically evaluated.	20–70y	Gastric band, Roux-en-Y gastric bypass (48 months)	Bariatric surgery had a negative effect on oral health, making the patient more susceptible to tooth erosion. The integration of dentists into teams essential for early diagnosis and treatment of lesions caused by tooth erosion. The awareness of dentists about the effects of Bariatric surgery in oral health can help them improve the quality of life of bariatric patients.
Sales-Peres, S. H.C.et al. 2015 (Brazil)	Cohort	50 patients undergoing bariatric surgery - 8 men and 42 women	38–49y	NR (6 and 12 months)	Periodontal disease increased in severity. Systemic inflammation due to bariatric surgery in obese individuals does not seem to affect the course of periodontal disease.
Sales-Peres, S.H.C. et al. 2016 (Brazil)	Cohort	110 morbidly obese patients	20–60y	Roux-en-Y gastric bypass (6 and 12 months)	The findings suggest that weight loss was associated with increased gingival bleeding, peaking at 6 months after bariatric surgery. Periodontal loss of insertion remained unchanged during the study period.
Schwenger, K.J.P.et al. 2020 (Canada)	Cohort	59 patients, of which only 29 completed the study.	47y	Roux-en-Y gastric bypass (1 and 6 months)	More than half of the patients had hyposalivation before gastric bypass and this was associated with higher fasting glycemia. Hyposalivation improved after gastric bypass in parallel with improvements in metabolic parameters, but there was no change in chronic oral inflammation.
Souza, A.C.D et al. 2013 (Brazil)	Case-control	62 patients - 31 were candidates for surgery bariatric surgery (control group) and 31 had already undergone Roux surgery. Gastric bypass surgery (case group).	18–65y	Roux-en-Y gastric bypass (17.7 months)	There is no significant association between halitosis and Rouxen-Sbypass gastric disease. However, they highlight the possible effect of this surgery on the oral cavity.
Tavares P.V. et al. 2015 (Brazil)	Cohort	20 volunteers	23–58y	NR (2 and 6 months)	Despite the beneficial effects of bariatric surgery on the general state of health, the negative effects reflect on the state of oral health and masticatory function performance. Therefore, it is necessary to integrate surgeon dentistry into the multidisciplinary team, monitoring these patients before and after surgery.

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

Table 1 (continued)

Author, Year (Country)	Study design	Sample	Age (Mean or range)	Type of surgery (Postoperative follow-up time)	Conclusions
Vargas, J.A. et al. 2019 (Brazil)	Cohort	31 women	20–35y	NR (6 months)	Obese patients present greater bone loss, which worsens after bariatric surgery when compared to eutrophic patients. The same happens with the bone trabeculate that becomes more sparse after the operation.
Weinberg G. et al. 2018 (Israel)	Cohort	50 patients	37,5	NR (12 months)	Bariatric surgery had no significant effect on oral health or on the perception of quality of life influenced by oral conditions, despite significant weight loss and improvement in general health parameters.
Yamashita, J.M. et al. 2017 (Brazil)	Cohort	150 patients, 75 obese, 75 eutrophic	18–65y	NR (6 months)	Individuals with metabolic syndrome, compared to eutrophic individuals, have a higher number of carious lesions before and after bariatric surgery. Bariatric surgery did not interfere in the progression of carious lesions.

Legends: NR—Not reported; y—Years old

[7, 35, 38–45], and three presented a high risk of bias [39, 46–48] (Appendix 3).

The factors considered as limitations of the studies addressed in this review included the deficient reports, such as the lack of detail in the inclusion and exclusion criteria; confounding factors; and the lack of control for these factors.

Results of individual studies

Chewing

Two studies addressed masticatory functions. According to Tavares et al. (2015), there was a worsening in the performance of the masticatory function when comparing pre- and postoperative periods, with malocclusion remaining during the study [34]. Likewise, Godlewski et al. (2011) evaluated changes in the masticatory function of patients that underwent BS and reported that all obese patients changed their masticatory kinematics regardless of dental status. However, the increase in chewing time depends on the type of food and the dental status of the individuals [24]. A change in muscle activity and bite force in the postoperative period was also observed in these patients [49].

Alveolar bone loss

Only one study addressed bone loss, which comprised 31 obese female patients with greater bone loss and decrease in trabecular bone volume after BS compared to eutrophic patients [47].

Halitosis

There was no unanimity regarding the change in halitosis in patients when comparing pre- and postoperative periods. Studies reported no impact of bariatric surgery on the rate of individuals with halitosis [34, 50]. On the other hand, according to Alzahrani et al. (2021), patients that underwent BS with a high decrease in body mass index (BMI) had an increase in ketone breath, consequently promoting an increase in halitosis [22].

Hyposalivation

There was controversy regarding salivary flow, as some studies reported decreased hyposalivation after BS [8, 27, 39, 48]. Patients who presented hyposalivation were associated with high fasting glucose [8]. Yamashita et al. (2017) found that salivary flow increased after BS [48]. Improvement in periodontal conditions were not observed, although the salivary flow increased after surgery [39].

Studies reported alterations in the salivary composition [25, 27, 29, 30, 39–41, 48]. Gambus et al. (2019) evaluated

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

sialometric and sialochemical changes in morbidly obese patients and concluded that both weight gain and BS altered the salivary composition [41]. Lamy et al. (2015) observed changes in the salivary proteome, which increased in patients that underwent bariatric surgery compared to obese individuals and also differed from the salivary proteome of those with normal weight [29].

Dental aspects

Regarding dental aspects, studies have shown that there is a worsening of tooth wear [7, 23, 30, 38, 39, 46], the appearance of carious lesions [30, 39, 43, 48, 50], and dental hypersensitivity in patients that underwent BS [34].

In the case of carious lesions, some studies reported an increase in lesions after surgery [30, 39, 43, 48]. However, individuals with metabolic syndrome have more carious lesions before and after BS than eutrophic individuals, and the surgery did not interfere with the progression of carious lesions [48].

Regarding tooth wear, patients who underwent BS had a higher rate of tooth wear on the incisal/occlusal surfaces, with anterior teeth being the most affected [23]. The metabolic action and effects of BC on oral health can increase the susceptibility of dental erosion [46]. Yang et al. (2021) observed that both the obese population and operated patients with vitamin D deficiency had a greater probability of developing erosive tooth wear[37].

Periodontal aspects

Some studies indicated a periodontal worsening in these patients [28, 32, 33, 44]. Moreover, there was an increase in periodontal diseases such as periodontal pockets[30, 43] and gingival bleeding [35], after surgery. On the other hand, Jaiswal et al. (2015) pointed out that diet modification and oral hygiene maintenance can help to improve periodontal aspects and the quality of life after surgery [26].

Results of syntheses

Fifteen studies were included in the quantitative synthesis. Random effects meta-analyses were performed to compare the mean baseline and post-surgical follow-up times for the following outcomes: decayed, missing, and filled teeth (DMFT); mean missing teeth; % tooth wear; salivary flow; and probing depth. Studies that did not evaluate at least one of these variables or which did not describe the data included in the meta-analysis were excluded from the quantitative synthesis.

When comparing the DMFT of morbidly obese patients in the preoperative period, there was no difference between times [DM = —0.39; CI95% = -0.91 – 0.14; $I^2 = 0\%$], with no heterogeneity between the results regardless of the postoperative evaluation time considered (which ranged between two and 16 months) (Fig. 2).

Likewise, there was no significant difference between the number of missing teeth when comparing pre- and postoperative periods [MD = 0.32; CI95% = -0.47 – 1.10; $I^2 = 0\%$], with no heterogeneity between the effect sizes of the studies also when considering periods of 3 years postoperation (ranging from three to 36 months) (Fig. 3).

Regarding the rate (%) of tooth wear, the analysis was subdivided into the rate of dentin wear, enamel wear, and unspecified wear. When considering the global effect, there was no difference between times [MD -2.08; CI95% = -5.65 – 1.49; $I^2 = 84.59\%$]. However, when the analysis was subdivided into the location of tooth wear, a decrease in the rate of enamel wear [MD = 3.29; CI95% = 1.39 – 5.20; $I^2 = 0\%$; Hedges' $g = 0.38$] and an increase in the rate of dentin wear [MD = -6.23; CI95% = -8.45—-4.01; $I^2 = 0\%$; Hedges' $g = 0.61$] were observed, being both statistically significant

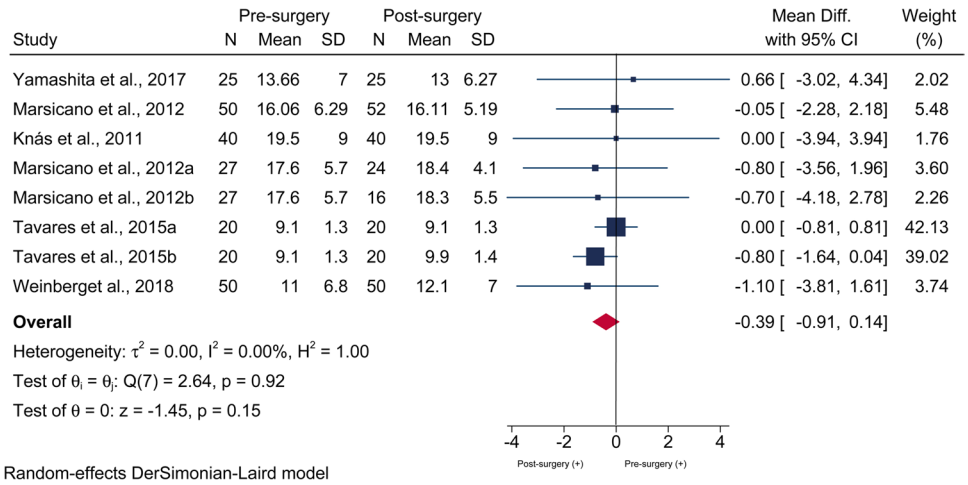

Fig. 2 Forest plot of the meta-analysis of DMFT, displaying the risk of bias judgments for each study included

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

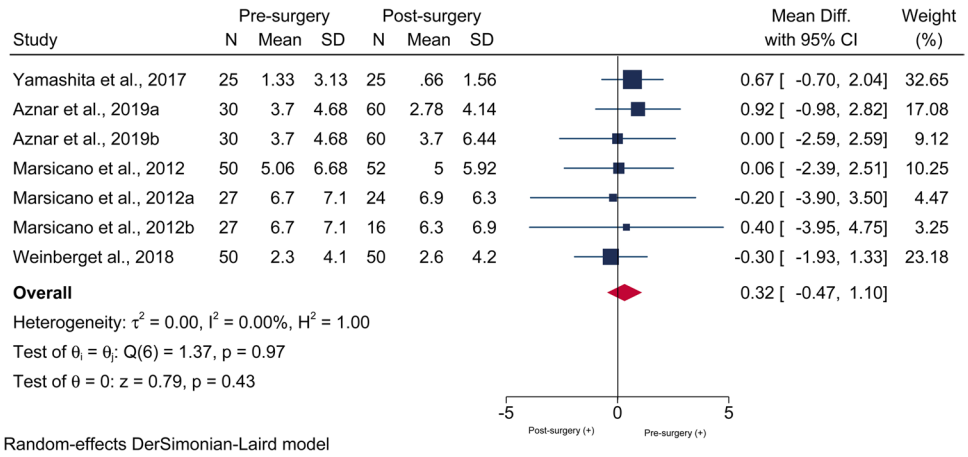

Fig. 3 Forest plot of the meta-analysis of the number of missing teeth, displaying the risk of bias judgments for each study included

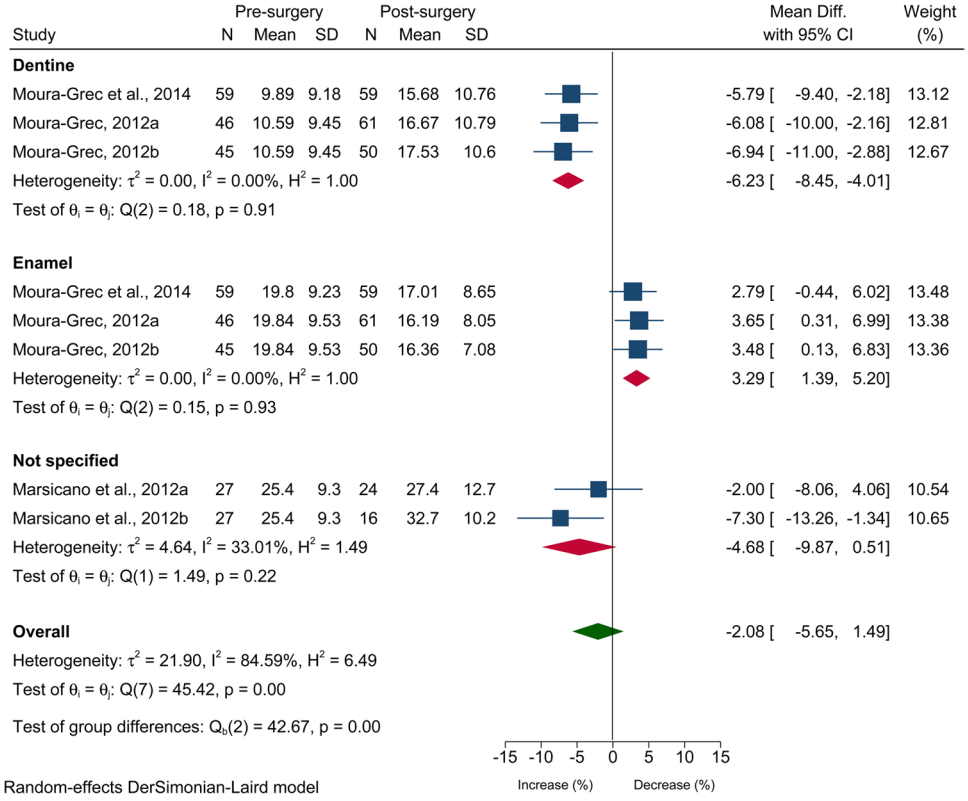

Fig. 4 Forest plot of the meta-analysis of % tooth wear, displaying the risk of bias judgments for each study included

in a postoperative evaluation period ranging from three to 12 months (Fig. 4).

Salivary flow also showed a statistical difference when comparing pre- and postoperative moments, with a slight mean increase (ml/min) in the postoperative period [MD = -0.04; CI95% = -0.09—-0.01; $I^2 = 3.53\%$]. However, despite the statistical significance, the upper limit of the interval was very close to the null line. Thus, the effect size was considered small (Hedges' $g = 0.17$) and not clinically significant (Fig. 5).

There was a statistically significant difference when considering probing depth between pre- and postoperative periods, with an increase in probing depth (mm) in the postoperative period [MD = -0.17; CI95% = -0.27—-0.07; $I^2 = 50.24\%$]. However, when the analysis was divided into subgroups considering the follow-up time, statistical significance was only observed for the 6-month postoperative period. Despite the difference between times, the effect size was considered small (Hedges' $g = 0.34$) (Fig. 6).

Reporting biases

The Egger test ($p > 0.05$) did not identify asymmetry of the funnel plot when considering the evaluation of salivary flow.

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
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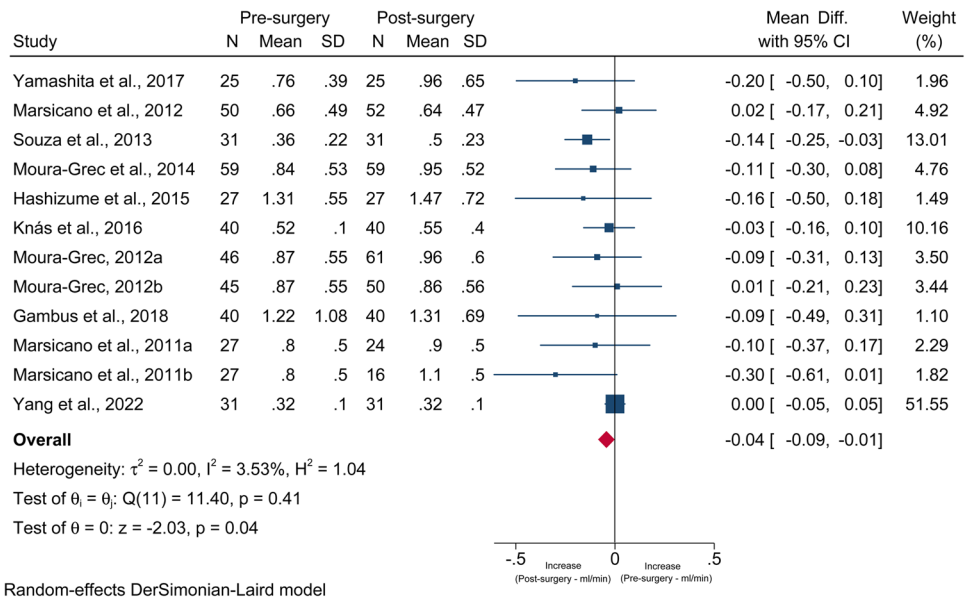

Fig. 5 - Forest plot of the meta-analysis of salivary flow, displaying the risk of bias judgments for each study included

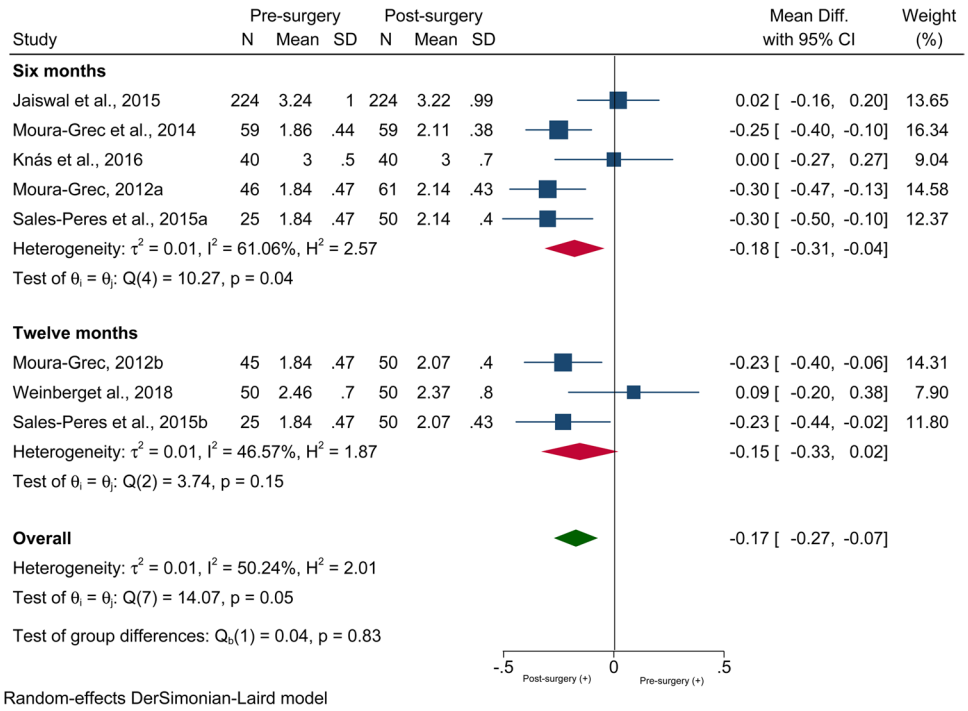

Fig. 6 Forest plot of the meta-analysis of probing depth, displaying the risk of bias judgments for each study included

The other outcomes could not be evaluated by the funnel plot and Egger test ($n < 10$).

Only one study with a high risk of bias was included in the quantitative synthesis [48]. When performing the sensitivity analysis without this study, there was little variation in the mean estimate and the respective confidence intervals, with the following estimates: DMFT [MD = 0.42; CI95% = -0.12 – 0.95; $I^2 = 0\%$]; Salivary flow [MD = -0.04; -0.08 – 0.01; $I^2 = 2.26\%$]; Number of missing teeth [MD = 0.15; CI95% = -0.81 – 1.10; $I^2 = 0\%$].

Certainty of evidence

The certainty of evidence was judged as low (DMFT, Number of missing teeth, and salivary flow) and very low (rate of tooth wear and probing depth). Furthermore, the certainty of evidence was reduced due to the observational design of the included studies, being prone to the influence of confounding factors and inconsistency for the outcomes 'rate of tooth wear' and 'probing depth' (Table 2).

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

Table 2 – Summary of findings table

Certainty assessment							№ of patients		Effect	Certainty
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Pre-surgery	Post-surgery	Mean Diff (95% CI)	
DMFT										
8	observational studies	not serious	not serious	not serious	not serious	none	259	247	-0.17 (-0.27—-0.07)	⊕⊕◯◯ Low
Number of missing teeth										
7	observational studies	not serious	not serious	not serious	not serious	none	239	287	0.32 (-0.47 – 1.10)	⊕⊕◯◯ Low
% of tooth wear										
8	observational studies	not serious	serious[a]	not serious	not serious	none	354	380	-2.08 (-5.65 – 1.49)	⊕◯◯◯ Very low
Salivary flow										
1 1	observational studies	not serious	not serious	not serious	not serious	none	448	456	-0.09 (-0.15—-0.03)	⊕⊕◯◯ Low
Probing depth										
8	observational studies	not serious	serious[b]	not serious	not serious	none	514	584	-0.09 (-0.15—-0.03)	⊕◯◯◯ Very low

CI: confidence interval

[a]Presence of heterogeneity, explored by subgroup analysis

[b]Unidentified source of heterogeneity

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

Discussion

This study aimed to evaluate the effects of BS on oral health through a systematic review based on the existing literature by quantitatively evaluating the following aspects: carious lesions, missing teeth, rate of tooth wear, salivary flow, and probing depth. Existing data demonstrate a higher rate of dentinal wear in these patients and a slight difference in salivary flow and probing depth.

The bariatric surgery comprises a set of techniques whose main objective is to promote weight reduction and disease regression [1]. Despite improving the self-esteem of post-surgery patients, some psychological and physiological changes can increase post-BS effects, which may reflect on the oral health condition. Different studies highlight the high predominance of oral manifestations in the pre- and post-operative periods of obese patients [4]. There was no difference in DMFT and the number of missing teeth between pre- and postoperative periods. However, most of the studies included in the analysis evaluated patients at six months postoperation. Thus, more follow-up time may be needed for the patient to present the outcome, especially when considering tooth loss.

Obesity is considered a risk factor for tooth wear regardless of the evaluation period (pre- or post-surgery) in bariatric patients [37]. According to Porcelli et al. (2016), most patients undergoing BS have induced vomiting (which can cause erosive tooth loss) and aggravating wear patterns that may already be present before surgery. Other studies also reported that approximately half of the patients undergoing surgery have gastrointestinal reflux disease and regurgitation [26, 44, 49]. The gastric acid makes the internal oral anatomical surfaces vulnerable to the proteolytic effects of acid, inducing erosive lesions in the oral mucosa, teeth, and periodontal tissues [25]. In addition, most bariatric patients reported to brush their teeth after vomiting. Thus, the combination between the decrease in pH in the oral cavity and the abrasion caused by the toothbrush contribute to the progression of tooth wear [46]. These data may explain the tendency of pre-existing wear to worsen, with a higher rate of wear at the dentin level in these postoperative patients, which was observed in this meta-analysis.

Some studies included in this review observed a decreased salivary flow after BS [8, 27, 39, 48]. The salivary flow rate can be altered by short-term complications after BS caused by dehydration [5, 51]. Furthermore, other factors that cause complications in the salivary flow rate include nutritional deficiencies [52] and the medications used by patients [5]. These factors are considered long-term complications [4]. However, despite the statistical significance observed in the meta-analysis, the effect size was considered small, and there may be little or no clinical effect in these patients.

Jaiswal et al. (2015) observed that patients undergoing BS showed a considerable improvement in periodontal health, as these patients underwent changes in their diets and were instructed to improve oral hygiene practices during the preparation for surgery [26]. On the other hand, studies described a possible association between obesity and periodontal disease. Several inflammatory markers, such as (CRP), pro-inflammatory cytokines, and alpha tumor factors potentiate gingival inflammation. Thus, an environment that is conducive to bacterial proliferation on dental surfaces is created, which in turn can spread periodontitis [2, 12]. This study observed an increase in probing depth in the 6-month postoperative period. However, this difference was considered small and may not have clinical relevance.

It is worth mentioning some of the limitations of this review, such as the follow-up time of the included studies. Most of the studies included in this review were characterized by a short-term follow-up of patients [26, 30, 33, 39, 44, 45]. A longer follow-up period in patients undergoing surgery can increase the certainty of the evidence generated, allowing these patients to reach a state of balance and completely react to the surgical impact [26, 30, 33, 39, 44, 45]. Moreover, it must be considered that tooth loss is a process that can take a long time to occur, and it is not possible to measure it with great certainty in a short time. Given the above in this review, it can be observed that although this procedure generates several benefits for the patients, greater attention regarding the oral health condition should be given to these patients, especially regarding dental wear. Patients undergoing surgery should receive careful monitoring by doctors, dentists, and the entire multidisciplinary team involved before and after the surgical procedure. Through this, patients can significantly improve their quality of life and reduce the incidence of the different types of injuries mentioned above [25, 30].

Conclusions

Patients undergoing bariatric surgery may show no difference when considering the rate of caries and tooth loss and little or no effect on salivary flow and periodontal probing depth compared to obese patients that did not undergo BS. On the other hand, greater attention should be given to dentin wear in post-surgical patients.

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Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

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Declarations

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Informed consent Not applicable.

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Competing interests The authors declare no competing interests.

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Azevedo ML, Silva NR, Mafra CA, Lins R, Dantas EM, Gurgel BC, Martins ARL (2020) Oral Health Implications of Bariatric Surgery in Morbidly Obese Patients: An Integrative Review. Obes Surg 30:1574–1579. https://doi.org/10.1007/s11695-019-04334-0
2. Dos Santos MCM, Pellizzer EP, SoutoMaior JR, Casado B, Gomes JM, Vasconcelos B, Moraes SLD (2019) Clinical periodontal conditions in individuals after bariatric surgery: a systematic review and meta-analysis. Surg Obes Relat Dis 15:1850–1859. https://doi.org/10.1016/j.soard.2019.07.008
3. World Health Organization (WHO) (2021) Obesidad y sobrepeso. https://www.who.int/es/news-room/fact-sheets/detail/obesity-and-overweight. Accessed 11 Nov 2021
4. Quintella MCM, Farias T, SoutoMaior JR, Casado B, Leao RS, Moraes SLD (2020) Relationship between bariatric surgery and dental erosion: a systematic review. Surg Obes Relat Dis 16:1283–1290. https://doi.org/10.1016/j.soard.2020.04.044
5. Taghat N, Mossberg K, Lingstrom P, Bjorkman S, Lehrkinder A, Werling M, Ostberg AL (2021) Oral health profile of postbariatric surgery individuals: A case series. Clin Exp Dent Res 7:811–818. https://doi.org/10.1002/cre2.411
6. Castilho AVSS, Foratori-Junior GA, Sales-Peres SHC (2019) Bariatric Surgery Impact on Gastroesophageal Reflux and Dental Wear: A Systematic Review. Arq Bras Cir Dig 32:e1466. https://doi.org/10.1590/0102-672020190001e1466
7. Grec PGM (2012) Impacto da cirurgia bariátrica na condição periodontal e quantificação de bactérias periodontopatogênicas por meio da q-PCR: estudo longitudinal. Tese de Doutorado, Faculdade de Odontologia de Bauru, Universidade de São Paulo, Bauru. https://doi.org/10.11606/T.25.2012.tde-12062012-103655
8. Schwenger KJP, Alghamdi MM, Ghorbani Y, Jackson TD, Okrainec A, Allard JP (2020) Hyposalivation is prevalent in bariatric patients but improves after surgery. Surg Obes Relat Dis 16:1407–1413. https://doi.org/10.1016/j.soard.2020.06.005
9. Taghat N, Werling M, Östberg AL (2020) Oral Health-Related Quality of Life After Gastric Bypass Surgery. Obes Surg 30:224–232. https://doi.org/10.1007/s11695-019-04172-0
10. Farias TMCP, Vasconcelos BCDE, SoutoMaior JR, Lemos CAA, Moraes SLD, Pellizzer EP (2019) Influence of Bariatric Surgery on Salivary Flow: a Systematic Review and Meta-Analysis. Obes Surg 29:1675–1680. https://doi.org/10.1007/s11695-019-03784-w
11. Colak D, Gaspersic R, Kucic AC, Pintar T, Gaspirc B (2021) The effect of bariatric surgery on periodontal health: systematic review and meta-analyses. Arch Med Sci 17:1118–1127. https://doi.org/10.5114/aoms/135880
12. Fontanille I, Boillot A, Range H, Carra MC, Sales-Peres SHC, Czernichow S, Bouchard P (2018) Bariatric surgery and periodontal status: A systematic review with meta-analysis. Surg Obes Relat Dis 14:1618–1631. https://doi.org/10.1016/j.soard.2018.07.017
13. Souza MSG, Oliveira WD, Lages SF, Fernandes AI, Falci GMS (2018) Relationship between bariatric surgery and periodontal status: a systematic review and meta-analysis. Surg Obes Relat Dis 14:1205–1216. https://doi.org/10.1016/j.soard.2018.04.018
14. Salgado-Peralvo AO, Mateos-Moreno MV, Arriba-Fuente L, Garcia-Sanchez A, Salgado-Garcia A, Peralvo-Garcia V, Millan-Yanes M (2018) Bariatric surgery as a risk factor in the development of dental caries: a systematic review. Public Health 155:26–34. https://doi.org/10.1016/j.puhe.2017.11.013
15. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hrobjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372:n71. https://doi.org/10.1136/bmj.n71
16. Wan X, Wang W, Liu J, Tong T (2014) Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. BMC Med Res Methodol 14:135
17. Moola S, Munn Z, Tufanaru C, Aromataris E, Sears K, Sfetcu R, Currie M, Qureshi R, Mattis P, Lisy K, Mu P-F (2020) Chapter 7: Systematic reviews of etiology and risk . In: Aromataris E, Munn Z (Editors). JBI Manual for Evidence Synthesis. JBI, 2020. Available from. https://synthesismanual.jbi.global
18. Basso IB, Goncalves FM, Martins AA, Schroder AGD, Taveira KVM, Stechman-Neto J, Santos RS, Guariza-Filho O, de Araujo CM (2022) Psychosocial changes in patients submitted to orthodontic surgery treatment: a systematic review and meta-analysis. Clin Oral Investig 26:2237–2251. https://doi.org/10.1007/s00784-021-04304-w
19. Hedges LV, Olkin I (1985) Statistical methods for meta-analysis. Academic Press, San Diego, CA
20 Andrade C (2020) Mean Difference, Standardized Mean Difference (SMD), and Their Use in Meta-Analysis: As Simple as It Gets. J Clin Psychiatry 81:11349. https://doi.org/10.4088/JCP.20f13681
21 Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, Schunemann HJ, Group GW (2008) GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ 336:924–6. https://doi.org/10.1136/bmj.39489.470347.AD

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

22. Alzahrani H, AlSarhan MA, Aldohayan A, Bamehriz F, Alzoman HA (2021) Extraoral Halitosis due to Exhaled Acetone in Patients Undergoing Sleeve Gastrectomy. JCDR. https://doi.org/10.7860/jcdr/2021/46260.14467
23. Aznar FD, Aznar FD, Lauris JR, Chaim EA, Cazzo E, Sales-Peres SHC (2019) Dental Wear and Tooth Loss in Morbid Obese Patients after Bariatric Surgery. Arq Bras Cir Dig 32:e1458. https://doi.org/10.1590/0102-672020190001e1458
24. Godlewski AE, Veyrune JL, Nicolas E, Ciangura CA, Chaussain CC, Czernichow S, Basdevant A, Hennequin M (2011) Effect of dental status on changes in mastication in patients with obesity following bariatric surgery. PLoS One 6:e22324. https://doi.org/10.1371/journal.pone.0022324
25. Hashizume LN, Bastos LF, Cardozo DD, Hilgert JB, Hugo FN, Stein AT, Souto KE, Meinhardt NG (2015) Impact of Bariatric Surgery on the Saliva of Patients with Morbid Obesity. Obes Surg 25:1550–1555. https://doi.org/10.1007/s11695-015-1741-4
26 Jaiswal GR, Jain VK, Dhodapkar SV, Kumathalli KI, Kumar R, Nemawat A, Jain A (2015) Impact of Bariatric Surgery and Diet Modification on Periodontal Status: A Six Month Cohort Study. J Clin Diagn Res 9:ZC43-5. https://doi.org/10.7860/JCDR/2015/14663.6489
27. Knas M, Maciejczyk M, Sawicka K, Hady HR, Niczyporuk M, Ladny JR, Matczuk J, Waszkiel D, Zendzian-Piotrowska M, Zalewska A (2016) Impact of morbid obesity and bariatric surgery on antioxidant/oxidant balance of the unstimulated and stimulated human saliva. J Oral Pathol Med 45:455–464. https://doi.org/10.1111/jop.12383
28. Lakkis D, Bissada NF, Saber A, Khaitan L, Palomo L, Narendran S, Al-Zahrani MS (2012) Response to periodontal therapy in patients who had weight loss after bariatric surgery and obese counterparts: a pilot study. J Periodontol 83:684–689. https://doi.org/10.1902/jop.2011.110230
29. Lamy E, Simoes C, Rodrigues L, Costa AR, Vitorino R, Amado F, Antunes C, do Carmo I (2015) Changes in the salivary protein profile of morbidly obese women either previously subjected to bariatric surgery or not. J Physiol Biochem 71:691–702. https://doi.org/10.1007/s13105-015-0434-8
30. Marsicano JA, Grec PG, Belarmino LB, Ceneviva R, Peres SH (2011) Interfaces between bariatric surgery and oral health: A longitudinal survey. Acta Cir Bras 26(Suppl 2):79–83. https://doi.org/10.1590/s0102-86502011000800015
31. Pataro AL, Cortelli SC, Abreu MH, Cortelli JR, Franco GC, Aquino DR, Cota LO, Costa FO (2016) Frequency of periodontal pathogens and Helicobacter pylori in the mouths and stomachs of obese individuals submitted to bariatric surgery: a cross-sectional study. J Appl Oral Sci 24:229–238. https://doi.org/10.1590/1678-775720150534
32. Pataro AL, Costa FO, Cortelli SC, Cortelli JR, Souza AC, Abreu MH, Girundi MG, Costa JE (2012) Influence of obesity and bariatric surgery on the periodontal condition. J Periodontol 83:257–266. https://doi.org/10.1902/jop.2011.100782
33. Sales-Peres SHC, Sales-Peres MC, Ceneviva R, Bernabe E (2017) Weight loss after bariatric surgery and periodontal changes: a 12-month prospective study. Surg Obes Relat Dis 13:637–642. https://doi.org/10.1016/j.soard.2016.08.007
34. Tavares PV, Coelho A, Vasconcelos MCR, Barros CMB, Coury RMMMSM, Nascimento AS, Correia MC, Fontes L (2015) Oral Health and Masticatory Performance in Adults Submitted to Bariatric Surgery: A Pilot Study. Open J Stomatol 05:87–93. https://doi.org/10.4236/ojst.2015.53013
35. Tinos AMFG, Foratori-Junior GA, Marcenes W, Camargo FB, Groppo FC, Sales-Peres SHC (2022) Impact of Bariatric Surgery in Anxiety and Oral Condition of Obese Individuals: A Cohort Prospective Study. Arq Bras Cir Dig 34:e1615. https://doi.org/10.1590/0102-672020210002e1615
36. Van Leeuwen SC, Nienhuijs SW, Pijpe J (2022) Oral health-related quality of life before and after bariatric surgery. Ned Tijdschr Tandheelkd 129:239–246. https://doi.org/10.5177/ntvt.2022.05.22006
37. Yang C, Hammer FJ, Reissfelder C, Otto M, Vassilev G (2021) Dental Erosion in Obese Patients before and after Bariatric Surgery: A Cross-Sectional Study. J Clin Med 10:4902. https://doi.org/10.3390/jcm10214902
38. AlvesMdo S, da Silva FA, Araújo SG, de Carvalho AC, Santos AM, de Carvalho AL (2012) Tooth Wear in Patients Submitted to Bariatric Surgery. Braz Dent J 23(2):160–166
39. de Moura-Grec PG, Yamashita JM, Marsicano JA, Ceneviva R, de Souza Leite CV, de Brito GB, Brienze SL, Sales-Peres SH (2014) Impact of bariatric surgery on oral health conditions: 6-months cohort study. Int Dent J 64:144–149. https://doi.org/10.1111/idj.12090
40. Dzunkova M, Liptak R, Vlkova B, Gardlik R, Cierny M, Moya A, Celec P (2020) Salivary microbiome composition changes after bariatric surgery. Sci Rep 10:20086. https://doi.org/10.1038/s41598-020-76991-6
41. Gambus LC, Johann AC, Bettega PV, Rosa EA, Morimoto IM, da Cruz MR, Nassif LS, Rocha SL, Loshii SO, Vaz AV, Solyon JA, Moraes F, Brancher JA, Ignácio SA, Hardy AM (2018) Sialometric and Sialochemical Changes in Morbidly Obese Patients before and after Bariatric Surgery. J Contemp Dent Pract 19(7):756–761. https://doi.org/10.5005/jp-journals-10024-2332
42. Malek-Zadeh CH, Watanabe LM, Junior WS, Vasconcelos PB, Regalo SCH, Nonino CB (2022) Evaluation of Stomatognathic System Parameters After Bariatric Surgery. Obes Surg 32:374–380. https://doi.org/10.1007/s11695-021-05774-3
43. Marsicano JA, Sales-Peres A, Ceneviva R, de C Sales-Peres SH, (2012) Evaluation of oral health status and salivary flow rate in obese patients after bariatric surgery. Eur J Dent 6(2):191–197
44. Sales-Peres SH, de Moura-Grec PG, Yamashita JM, Torres EA, Dionisio TJ, Leite CV, Sales-Peres A, Ceneviva R (2015) Periodontal status and pathogenic bacteria after gastric bypass: a cohort study. J Clin Periodontol 42:530–536. https://doi.org/10.1111/jcpe.12410
45. Weinberg GB, Bilder L, Horwitz J, Pupko M, Mahajna A (2018) Oral health status of patients before and after bariatric bariatric surgery. Dent Oral Health Care 1(1):1–8
46. Porcelli ICS, Roma CC, Nunes MCP, Maciel SM, Pascotto RC (2016) Effects of bariatric surgery on the oral health of patients. Int J Dent Oral Health 2(2):e1–6. https://doi.org/10.16966/2378-7090.181
47. Vargas JA, Bonato RCS, Orenha ES, Sales-Peres SHC (2020) Assessment of Alveolar Bone Pattern in Obese and Non-Obese Women, before and after Bariatric Surgery: A Prospective Cohort Study. Arq Bras Cir Dig 33:e1501. https://doi.org/10.1590/0102-672020190001e1501
48. Yamashita JM (2017) Avaliação de lesões cariosas e biofilme dentário por fluorescência em pacientes portadores da síndrome metabólica submetidos à cirurgia bariátrica: um estudo de coorte prospectivo. Tese de Doutorado, Faculdade de Odontologia de Bauru, Universidade de São Paulo, Bauru. https://doi.org/10.11606/T.25.2018.tde-20022018-154352
49. Raj PP, Bhattacharya S, Misra S, Kumar SS, Khan MJ, Gunasekaran SC, Palanivelu C (2019) Gastroesophageal reflux-related physiologic changes after sleeve gastrectomy and Roux-en-Y gastric bypass: a prospective comparative study. Surg Obes Relat Dis 15:1261–1269. https://doi.org/10.1016/j.soard.2019.05.017
50. Souza AC, Franco CF, Pataro AL, Guerra T, Costa F, da Costa JE (2013) Halitosis in obese patients and those undergoing bariatric

Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11

surgery. Surg Obes Relat Dis 9:315–321. https://doi.org/10.1016/j.soard.2011.10.020
51. Ivanics T, Nasser H, Leonard-Murali S, Genaw J (2019) Dehydration risk factors and impact after bariatric surgery: an analysis using a national database. Surg Obes Relat Dis 15:2066–2074
52. Schulman AR, Thompson CC (2017) Complications of bariatric surgery: what you can expect to see in your GI practice. Am J Gastroenterol 112(11):1640–1655. https://doi.org/10.1038/ajg.2017.241

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Guilherme Ricardo Cabral dos S
guilhermercs@yahoo.com.br
115.350.439-11